

NOVA PAY

Machine-Learning Fraud Detection

Executive Report — Validated Results, Insights, Cost & Recommendations

Prepared by Dr. Ogheneworo Offeh
Audience: Executive stakeholders · Classification: Confidential

Executive Summary

Nova Pay's current defence against fraudulent money transfers relies on a hand-written, rule-based engine. Rules are static, age quickly, and struggle in a setting where fraud is rare — about **8.9% of transactions** — fast-moving, and costly. This report presents a validated machine-learning (ML) alternative: an end-to-end pipeline that learns fraud patterns from data, explains every decision, scores transactions in milliseconds, and renews itself as fraud evolves.

On a held-out test set of 2,240 transactions that the model never trained on, the tuned champion model (XGBoost) **catches 83% of fraud** while keeping **81% of its alerts correct** and a **false-alarm rate of just 1.9%**. Our headline recommendation is to approve a **shadow deployment** — running the ML model in parallel with current rules — to confirm this uplift on live traffic at zero customer risk.

Validated results at a glance

Metric	Result	What it means
Recall (fraud caught)	83.4%	166 of 199 frauds detected
Precision (flag quality)	81.4%	166 of 204 alerts were genuine fraud
False-alarm rate	1.9%	only 38 of 2,041 legitimate transactions blocked
PR-AUC	≈0.92	strong ranking power under heavy class imbalance
ROC-AUC	≈0.97–0.98	consistently high across all four models tested

***Note:** these figures are taken from the project's validation outputs at the model's operating point. The decision threshold can be tuned (see Section 2) to shift the balance between fraud caught and false alarms according to business appetite.*

1. Project Overview

The problem we are solving

Digital money-transfer fraud has three properties that defeat rule-based systems. It is *rare* (a severe class imbalance, so naive accuracy is meaningless); it is *adaptive* (fraudsters change tactics faster than rules can be rewritten); and it is *expensive in both directions* — missed fraud is a direct loss, while false alarms block legitimate customers and overload investigators. The objective is to catch more fraud, raise fewer false alarms, explain each decision, and adapt automatically.

The solution: an eight-stage pipeline

The project follows a disciplined, eight-stage workflow in which each stage is engineered for robustness and hands off cleanly to the next:

Stage	What it does	Why it matters
1. Collect & Profile	Establishes baseline fraud prevalence and missingness	Honest expectations for an imbalanced problem
2. Clean & Prepare	KNN imputation; outlier capping at the 99th percentile	Stable inputs; preserves the real distribution
3. Engineer Features	Velocity, mismatch, device, and behaviour features	Turns raw data into signals a model can learn
4. Explore (EDA)	Patterns by channel, geography, time; anomalies	Confirms what separates fraud from legitimate
5. Model & Tune	LogReg & RF baselines, then XGBoost & LightGBM, tuned	Selects the strongest model objectively
6. Validate & Explain	Untouched holdout; SHAP transaction-level reasoning	Honest performance, defensible decisions
7. Deploy	FastAPI /score endpoint, containerised with Docker	Portable, fast, production-ready scoring
8. Monitor & Improve	Drift detection and a quarterly retraining playbook	Keeps accuracy high as fraud evolves

2. Core Decisions & Insights

Several decisions are the reason the solution is robust and trustworthy. The exploratory analysis also surfaced clear, actionable fraud patterns.

What actually drives fraud

The strongest predictors of fraud are **behavioural**: transaction velocity over 1-hour and 24-hour windows tops the list, followed by internal and IP risk scores, chargeback history, and device trust. These are features engineered from data the rules engine never combines — and precisely the signals we preserved by setting raw identifiers aside.

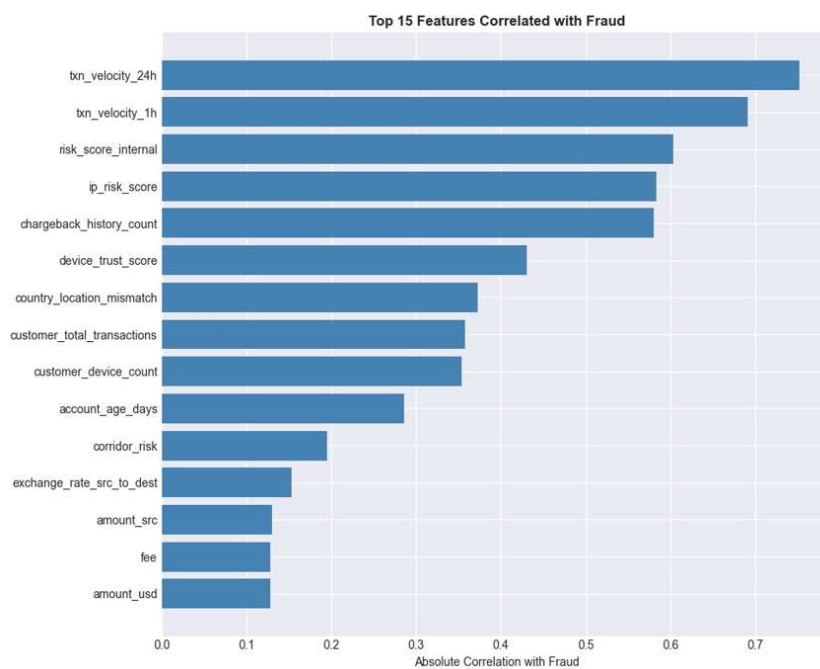


Figure 1. Top features correlated with fraud — transaction velocity dominates.

Where fraud concentrates: location mismatch

Geography is highly informative. Transactions where the customer's country and location **do not match** carry a fraud rate near **30%**, against roughly 3.5% when they agree — an eight-fold difference. Notably, volume and rate diverge: the US has the most fraud by count, while Canada and the UK show the highest fraud rates. A flat rule treating all geographies alike misses this entirely.

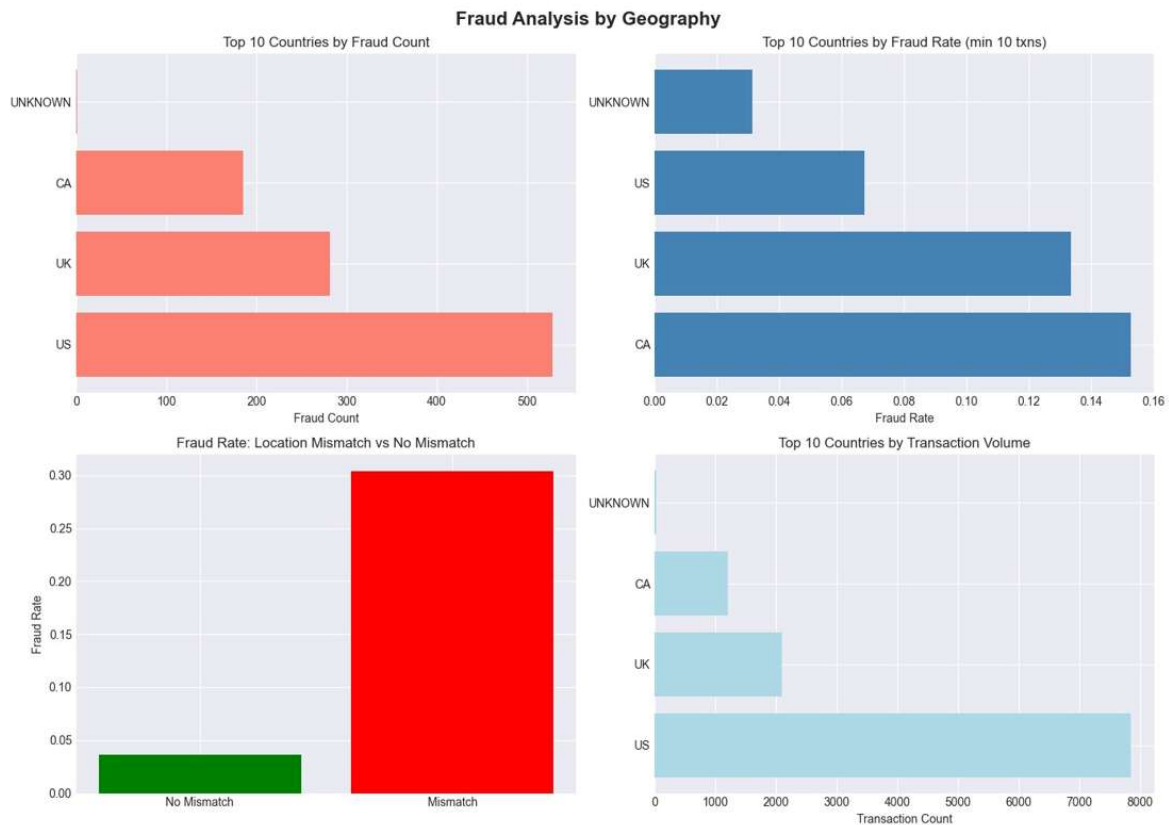


Figure 2. Fraud by geography — the location-mismatch panel (lower left) is the standout signal.

The exploratory analysis surfaced two further patterns: the **web channel** carries the highest fraud rate (~16%) despite lower volume than mobile, and fraud **spikes in the 4–7am window** across every day of the week — both consistent with automated or account-takeover activity.

Engineering decisions that make it robust

- **Separating identifiers from features.** Customer, device, and IP identifiers are kept aside and never fed to the model, preventing an encoding blow-up (on the order of 25,000 useless columns), data leakage, and overfitting to past fraudsters. Identities are converted into behaviour features instead.
- **Handling imbalance honestly.** With fraud at ~8.9%, a model that predicts “never fraud” would be ~91% accurate while catching nothing. We use stratified splits and class weighting, and judge on recall, precision and PR-AUC — never raw accuracy.
- **Building in explainability.** Every prediction carries SHAP values that show the features driving it — auditable by investigators and regulators.

Model comparison — gradient boosting wins

Four models were compared with cross-validation. All achieve high ROC-AUC (~0.97–0.98); the gradient-boosted models (XGBoost, LightGBM) give the best precision–recall balance under imbalance, and tuned XGBoost was selected as the champion.



Figure 3. Cross-validated model comparison across five metrics.

Champion performance and the business dial

On the held-out test set the tuned XGBoost model caught **166 frauds**, missed **33**, and raised only **38 false alarms**. The decision threshold is a business lever: raising it reduces false alarms, lowering it catches more fraud. The F1-optimal point sits near 0.85, but the right setting is a risk-appetite decision for the business.

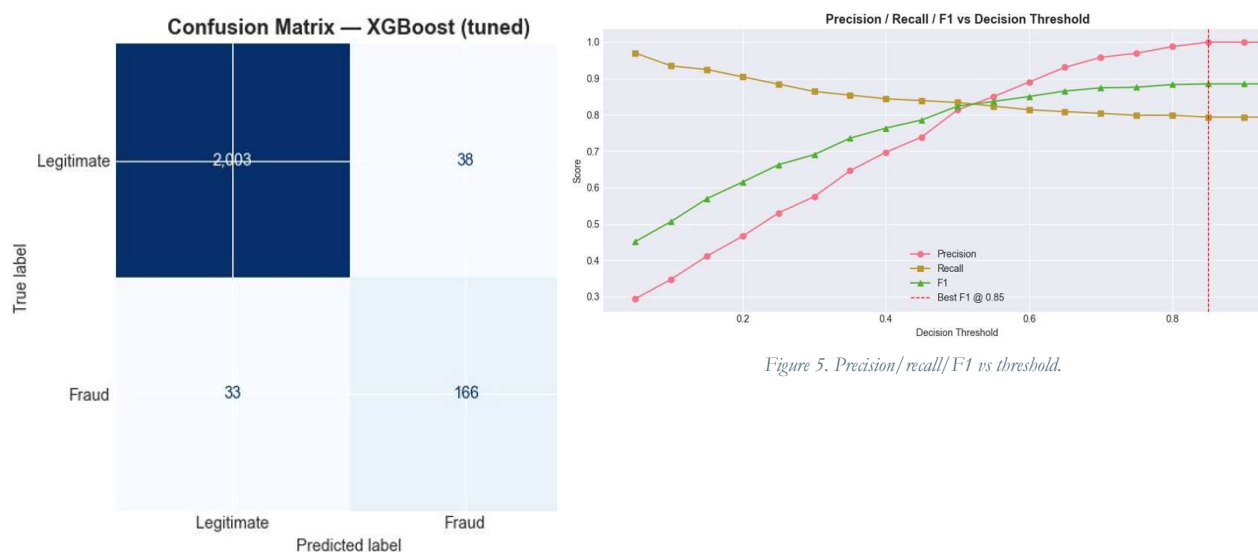


Figure 5. Precision/recall/F1 vs threshold.

Every decision is explainable (SHAP)

SHAP analysis provides both a global view of what drives fraud risk — led by **device fingerprint risk**, customer transaction history, IP risk and velocity — and a per-transaction breakdown showing exactly why any single transfer was flagged. This turns the model from a black box into an auditable decision aid.

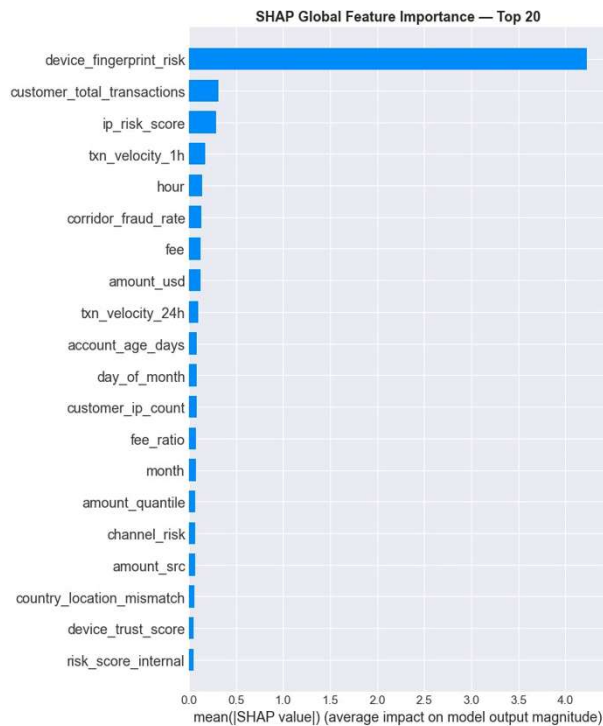


Figure 6. Global feature importance.

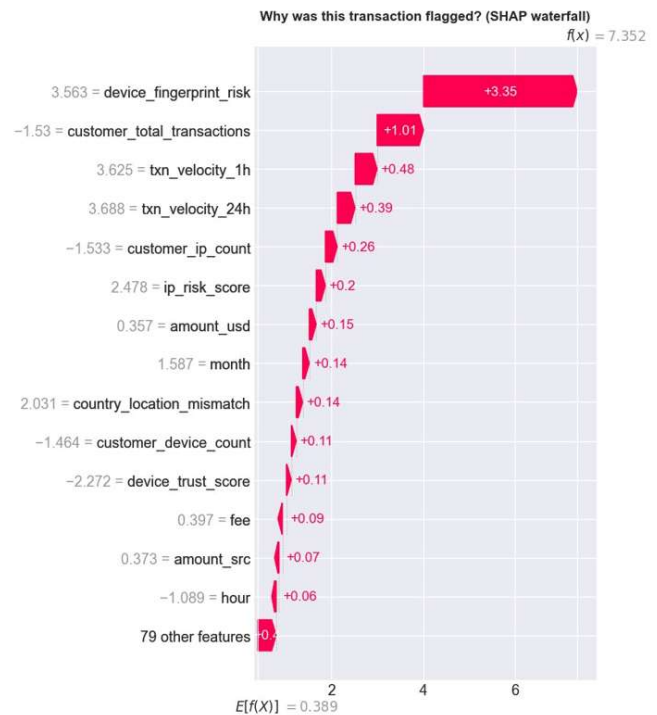


Figure 7. Why one transaction was flagged.

Why ML is more efficient than the rule-based engine

Capability	Rule-based engine	ML solution
Fraud caught	Limited to known, hand-coded patterns	83% recall, learning across many features
False-alarm control	Blunt thresholds; high false positives	1.9% false-alarm rate; tunable threshold
Adapting to new fraud	Manual rewrites, slow	Retraining on fresh data, automated
Explainability	Rule traceable but rigid	SHAP reasoning per transaction
Analyst efficiency	Flat alert lists	Risk-ranked queue — work the riskiest first

3. Cost Implications

The cost profile is a **front-loaded one-off build** followed by a **modest, predictable recurring run cost**. The data, modelling and validation work is already complete. The figures below are cost *types and drivers* for budgeting; precise amounts depend on infrastructure choices and internal rates, to be finalised with Finance and IT.

Phase	Activity	Cost type	Status
Data & modelling	Cleaning, EDA, feature engineering, modelling and tuning	One-off	Complete
Validation	Holdout testing, SHAP explainability, sign-off	One-off	Complete
Deployment	FastAPI scoring service, Docker, CI/CD, integration	One-off	Planned
Infrastructure	Compute and hosting for the scoring API	Recurring	Planned
Monitoring	Drift detection, dashboards, alerting	Recurring	Planned

Phase	Activity	Cost type	Status
Retraining	Quarterly model refresh as fraud evolves	Recurring	Planned

Cost offset: the build cost is substantially de-risked because the hard work is done. On the return side, at 83% recall and a 1.9% false-alarm rate, every prevented fraud and every avoided false block returns recovered losses and analyst hours. The recurring cost buys something the rule engine cannot — a model that *improves over time* rather than decaying.

Short, medium, and long-term value

Horizon	Value delivered
Short term (0–3 months)	Deploy scoring in parallel with current rules; immediate lift (~83% fraud caught at a 1.9% false-alarm rate); risk-ranked queue for analysts.
Medium term (3–12 months)	Retire brittle rules as the model proves out; tune the threshold to risk appetite; drift monitoring stabilises performance.
Long term (12+ months)	A self-renewing model via quarterly retraining; extensible to new products and corridors; a compounding data advantage over fraudsters.

4. Recommendations

We recommend the following sequence to move from a validated model to a value-generating production system, ordered to minimise risk:

- **Approve shadow deployment.** Run the ML model alongside current rules on live traffic to confirm the 83% / 1.9% result with zero customer impact.
- **Agree the operating threshold.** Set the fraud-caught versus false-alarm trade-off with Risk and Operations; the F1-optimal point sits near 0.85.
- **Stand up monitoring first.** Put drift detection and dashboards in place before full cutover so performance is observable from day one.
- **Invest in richer features.** Add graph/network signals and real-time velocity for the next step-change in accuracy.
- **Close the analyst feedback loop.** Feed investigator outcomes back into retraining so the model sharpens continuously.
- **Plan multi-product rollout.** Extend the same pipeline to new corridors and products once the first deployment is validated.

Recommended decision

Approve shadow deployment and the monitoring setup. This confirms the validated uplift (83% of fraud caught at a 1.9% false-alarm rate) on live traffic at minimal cost and zero customer risk, producing the evidence for a confident full rollout.

***Note on figures:** performance metrics and charts are the validation outputs of the Nova Pay project notebooks (held-out test set, $n=2,240$). Cost amounts are indicative planning aids to be confirmed with Finance and IT.*